

Integration Bee questions

Round 1

$$(1) \quad \int_0^{\pi^2} \cos(\sqrt{x}) dx = -4$$

Hint: sub $u = \sqrt{x}$

$$(2) \quad \int \tan^3(x) \sec^5(x) dx = \frac{\sec^7(x)}{7} - \frac{\sec^5(x)}{5}$$

Hint: $\tan^2(x) = \sec^2(x) - 1$

$$(3) \quad \int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \ln(e^x + e^{-x})$$

Hint: what is the derivative of the denominator?

$$(4) \quad \int x^2 \sin^2(x) dx = \frac{1}{24} (4x^3 - 6x \cos(2x) + (3 - 6x^2) \sin(2x))$$

Hint: $\cos(2x) = 1 - 2\sin^2 x$, solve for $\sin^2(x)$ and sub.

$$(5) \quad \int \frac{1}{2x^2 + 3} = \frac{1}{3} \ln \left| \frac{x}{2x+3} \right|$$

Hint: partial fractions

$$(6) \quad \int \frac{1}{x + x^e} dx = -\frac{\ln(x^{1-e} + 1)}{e - 1}$$

Hint: factor out x^e and then sub $u = x^{1-e} + 1$.

$$(7) \quad \int \frac{1}{\sin^2(x) + 2024} dx = \frac{1}{\sqrt{2025 \cdot 2024}} \arctan \left(\frac{\sqrt{2025x}}{\sqrt{2024}} \right)$$

Hint: $\sin(x) = \frac{\tan(x)}{\sec(x)}$

$$(8) \quad \int \frac{x^2 + x + 1}{x + 1} dx = \ln|x + 1| + \frac{x^2}{2}$$

Hint: divide out the polynomials then partial fractions

$$(9) \quad \int \frac{1}{1 + e^{-x}} dx = \ln(e^x + 1)$$

Hint: multiply top and bottom by e^x

$$(10) \quad \int e^{\arcsin x} dx = \frac{1}{2} (\sqrt{1 - x^2} + x) e^{\arcsin x}$$

Hint: sub $u = \arcsin x$ and use a trig identity

$$(11) \quad \int \frac{e^x}{e^{2x} + 1} dx = \arctan(e^x)$$

Hint: $u = e^x$

$$(12) \quad \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{\sin(x)} dx = \ln(\sqrt{3})$$

Hint: $\frac{1}{\sin(x)} = \frac{\sin(x)}{1 - \cos^2(x)}$

Round 1: Reserve

$$(13) \quad \int \frac{e^x}{e^{2x} + 2e^x + 1} dx = -\frac{1}{e^x + 1}$$

Hint: sub $u = e^x$

$$(14) \quad \int \frac{\sin\left(\frac{27}{x}\right)}{x^2} dx = \frac{\cos\left(\frac{27}{x}\right)}{27}$$

Hint: u sub

$$(15) \quad \int e^{e^x + x} dx = e^{e^x}$$

Hint: $e^{e^x + x} = e^{e^x} e^x$

$$(16) \quad \int \frac{2024^{\ln(x)}}{x} dx = \frac{2024^{\ln(x)}}{\ln(2024)}$$

Hint: sub $u = \ln(x)$

$$(17) \quad \int \frac{x}{\sqrt{1 + x^2}} dx = \sqrt{1 + x^2}$$

Hint: sub $u = 1 + x^2$

Round 2

$$(18) \quad \int \sqrt{\frac{1-x}{1+x}} dx = \arcsin x + \sqrt{1-x^2}$$

Hint: multiply by $1 = \sqrt{\frac{1-x}{1-x}}$.

$$(19) \quad \int \frac{x^{\frac{3}{4}} x^{\frac{5}{6}} x^{\frac{1}{2}} \dots}{\sqrt{x} \sqrt[4]{x} \sqrt[6]{x} \dots} dx = \frac{x^{1+\ln 2}}{1+\ln 2}$$

Hint: the Taylor series expansion for $\ln x$ gives $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

$$(20) \quad \int_0^{2024} x^{2024} \log_{2024}(x) dx = \frac{(2024)^{2025}}{2025} \left(1 - \frac{1}{\ln(2024)2025} \right)$$

Hint: use log conversion formula to change $\log_{2024}(x)$ into $\ln(x)$, then sub $u = \ln(x)$

$$(21) \quad \int \frac{1}{e^{\frac{x}{2}} + e^x} dx = 2(\ln|e^{-\frac{x}{2}} + 1| - 1)$$

Hint: sub $u = e^{-\frac{x}{2}}$

$$(22) \quad \int \frac{x^2}{\sqrt{1-x^2}} dx = \frac{1}{2}(\arcsin x - x\sqrt{1-x^2})$$

Hint: trig substitution

$$(23) \quad \int \frac{1}{\sqrt{x\sqrt{x\sqrt{x}} - x}} dx = 8 \arcsin \left(x^{\frac{1}{8}} \right)$$

Hint: $\sqrt{x\sqrt{x\sqrt{x}}} = x^{\frac{7}{4}}$

Round 2: Reserve

$$(24) \quad \int 3^x 3^{3^x} 3^{3^{3^x}} dx = \frac{1}{(\ln 3)^3} 3^{3^{3^x}}$$

Hint: $u = 3^{3^{3^x}}$

$$(25) \quad \int \sin^5(x) dx = -\frac{\cos^5(x)}{5} + \frac{2\cos^3(x)}{3} - \cos(x)$$

Hint: $\sin^4(x) = (1 - \cos^2(x))^2$

Final round

$$(26) \quad \int \sqrt{\left(\sin(20x) + 3\sin(21x) + \sin(22x) \right)^2 + \left(\cos(20x) + 3\cos(21x) + \cos(22x) \right)^2} dx = 3x + 2\sin(x)$$

Hint: expand out everything, then use trig identities (particularly angle sum formula) to simplify and then complete the square.

$$(27) \quad \int_0^{\frac{\pi}{2}} \frac{e^{\sec(x)}}{e^{\sec(x)} + e^{\csc(x)}} dx = \frac{\pi}{4}$$

Hint: sub $u = \frac{\pi}{2} - x$ and note what happens to the bounds.

$$(28) \quad \int \ln(x^2 + 13x + 40) dx = x \ln(x^2 + 13x + 40) - 2x + 8 \ln|x+8| + 5 \ln|x+5|$$

Hint: integrate by parts with $du = 1 \cdot dx$. Then divide out an x in the numerator and do partial fractions.

Round 3: Reserve

$$(29) \quad \int \frac{1}{x^3 - 1} dx = -\frac{1}{6} \left(\ln(x^2 + x + 1) + 2\sqrt{3} \arctan \left(\frac{2x+1}{\sqrt{3}} \right) - 2 \ln|2x+1| \right)$$

Hint: difference of cubes

$$(30) \quad \int \left(\frac{\ln x - 1}{1 + (\ln x)^2} \right)^2 dx = \frac{x}{(\ln x)^2 + 1}$$

Hint: sub $u = \ln(x)$ then expand the numerator

Note for presentation

In the venue, there are two boards on either side of the front end of the lecture theatre. Participants compete with each other two at a time, and each is assigned one board randomly. Each participant is also assigned a pair of judges who will focus on them.

Each pair of participants should be given 1 integral to solve at a time. The first to complete an integral correctly wins 1 point. A participant may indicate that they have found a solution to an integral by writing their final answer enclosed in a box. Once any participant presents a solution, the timer stops and the judges check the presented integral. A solution will be deemed to be correct if both judges assigned to them come to a consensus on the correctness of the answer. When a final answer is presented, the other participant may continue while judges come to a consensus regardless of whether the presented solution is correct or not. If the solution is incorrect, the timer continues again and the participant (for whom the timer was first stopped) may continue trying to find the right answer. Notice that presenting your final answer for judging, gives your opponent “free” time; a participant may not write while judging of their solution occurs. Hence, don’t just cycle a bunch of random answers. We’re capping the total writing time per integral (see later for more details).

We can announce that they do not need to include constants of integration for indefinite integrals. They must give the final answer in terms of the original variable.

Judges will be lenient about absolute values in solutions involving $\ln(x)$, although nothing will be announced about this explicitly.

The participant who receives the best score after 3 integrals, wins that match. In case of a tie (either by exact same completion time, or neither being able to finish after a long enough time), the participants are given a reserve integral. These are significantly easier than the other integrals in that round and will act as a “sudden death” tie-breaker.

A designated “observer” must observe both participants to know who finished first (in case it is very close), as judging each solution may take more or less time, but that should not affect who gets the point for fastest correct answer.

Order of integrals: The non-reserve integrals may be presented in random order to the participants. The reserve integrals are listed in descending order of difficulty. Present the reserve integrals in the exact order that they are listed here.

For each match, we first give the 3 integrals before going to reserves. So if neither solve integral no. 2, they first do no. 3 and then the reserve to determine who gets 2 points.

Time limits: There will be a 4 minute time limit for

integrals in round 1, a 5 minute time limit for round 2 and a 6 minute time limit for the final round.

Hints: A hint may be presented if both participants agree that they want one.